

First-Order Circuits

Video 11

Electrical Science

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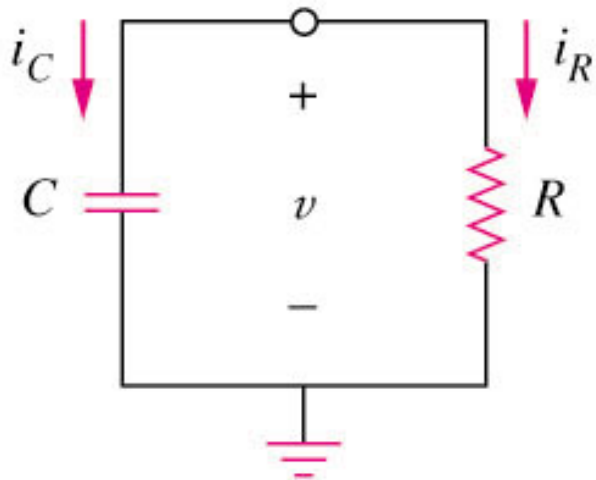
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First-Order Circuits

- The Source-Free RC Circuit
- The Source-Free RL Circuit
- Unit-step Function
- Step Response of an RC Circuit
- Step Response of an RL Circuit

The Source-Free RC Circuit

- A **first-order circuit** is characterized by a first-order differential equation.



By KCL

$$i_R + i_C = 0$$

$$\rightarrow \frac{v}{R} + C \frac{dv}{dt} = 0$$

$$v(t) = V_0 e^{-t/\tau}$$

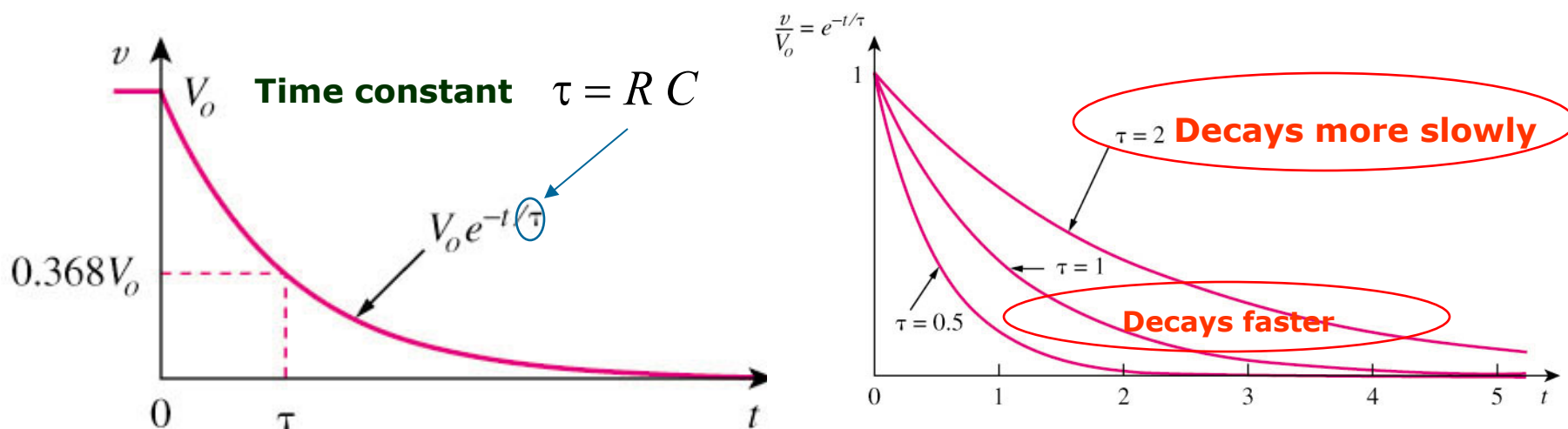
where

$$\tau = R C$$

- The initial voltage $v(0) = V_0$ across the capacitor.
- The time constant $\tau = RC$.

The Source-Free RC Circuit

- The **natural response** of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

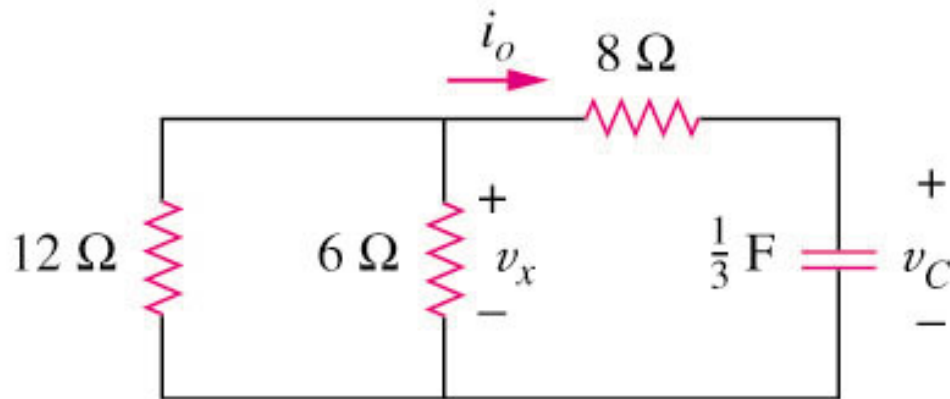


- The **time constant** τ of a circuit is the time required for the response to decay by a factor of $1/e$ or 36.8% of its initial value.
- v decays **faster** for small τ and **slower** for large τ .

The Source-Free RC Circuit

Example 1

Refer to the circuit below, determine v_C , v_x , and i_o for $t \geq 0$. Assume that $v_C(0) = 30$ V.

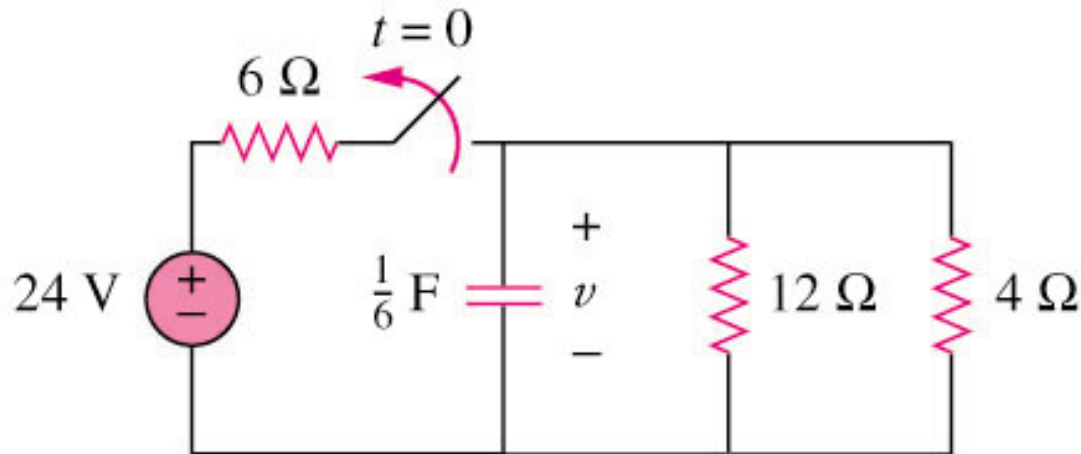


Answer: $v_C = 30e^{-0.25t}$ V ; $v_x = 10e^{-0.25t}$; $i_o = -2.5e^{-0.25t}$ A

The Source-Free RC Circuit

Example 2

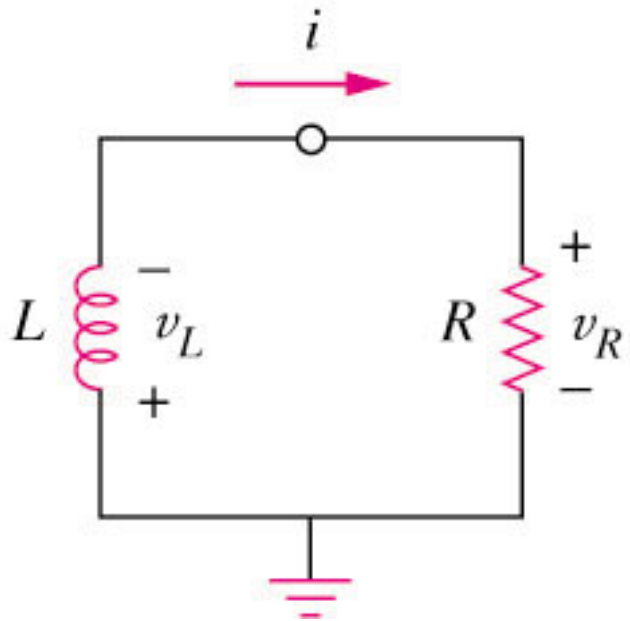
The switch in circuit below is opened at $t = 0$, find $v(t)$ for $t \geq 0$.



Answer: $V(t) = 8e^{-2t}\text{ V}$

The Source-Free RL Circuit

- A **first-order RL circuit** consists of a inductor L (or its equivalent) and a resistor (or its equivalent)



By KVL

$$v_L + v_R = 0$$

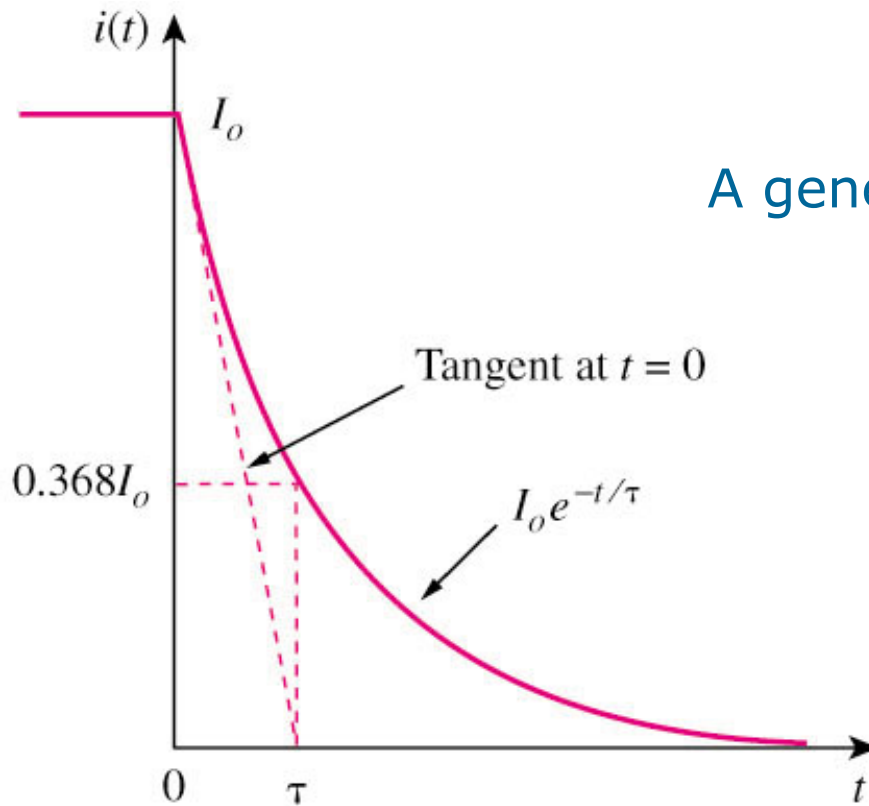
$$L \frac{di}{dt} + iR = 0$$



$$\frac{di}{i} = -\frac{R}{L} dt$$

$$i(t) = I_0 e^{-Rt/L}$$

The Source-Free RL Circuit



A general form representing a RL

$$i(t) = I_0 e^{-t/\tau}$$

where $\tau = \frac{L}{R}$

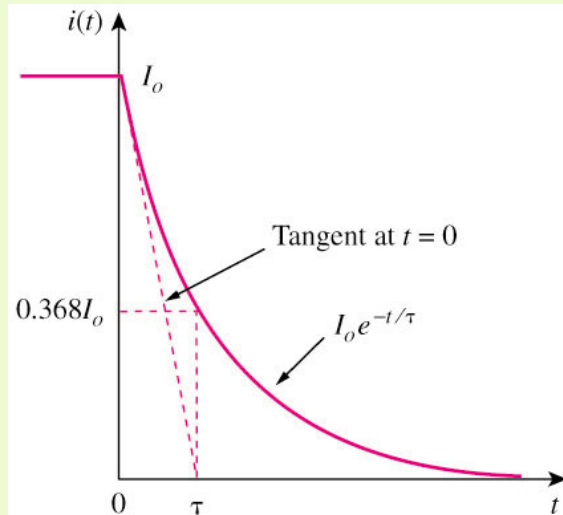
- The **time constant** τ of a circuit is the time required for the response to decay by a factor of $1/e$ or 36.8% of its initial value.
- $i(t)$ decays **faster for small τ** and **slower for large τ** .
- The general form is very similar to a RC source-free circuit.

The Source-Free RL Circuit

Comparison between a RL and RC circuit

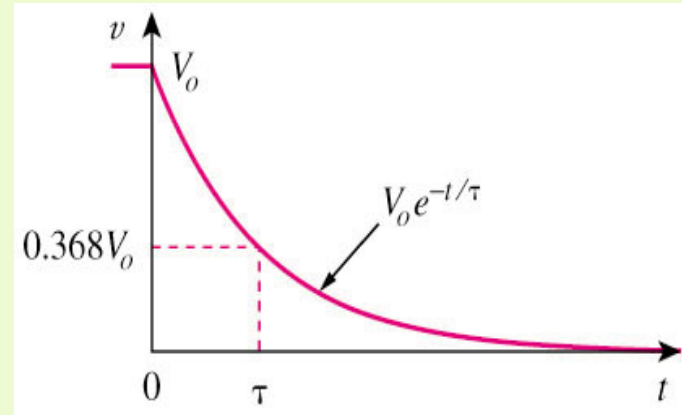
A RL source-free circuit

$$i(t) = I_0 e^{-t/\tau} \quad \text{where} \quad \tau = \frac{L}{R}$$



A RC source-free circuit

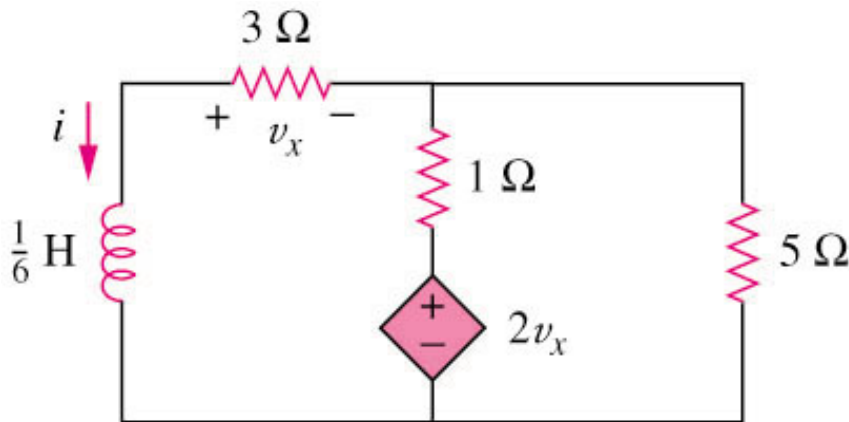
$$v(t) = V_0 e^{-t/\tau} \quad \text{where} \quad \tau = RC$$



The Source-Free RL Circuit

Example 3

Find i and v_x in the circuit. Assume that $i(0) = 5 \text{ A}$.



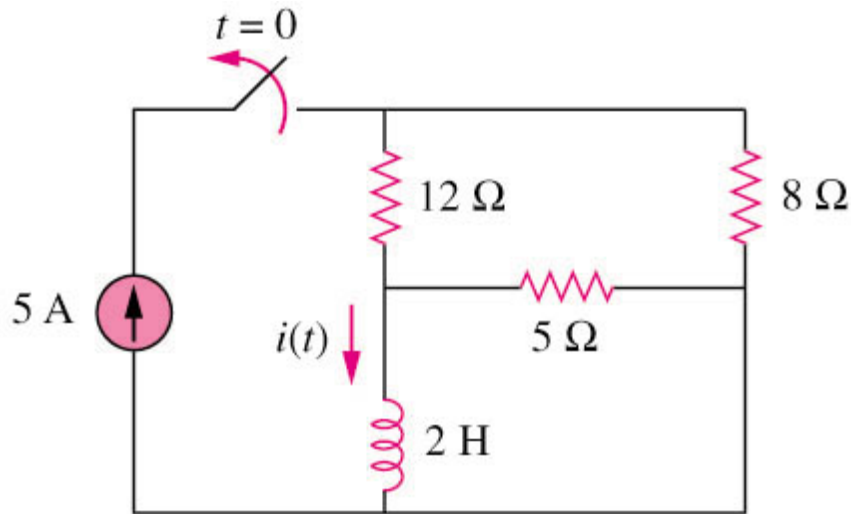
Answer: $i(t) = 5e^{-53t} \text{ A}$

$v_x(t) = -15e^{-53t} \text{ A}$

The Source-Free RL Circuit

Example 4

For the circuit, find $i(t)$ for $t > 0$.

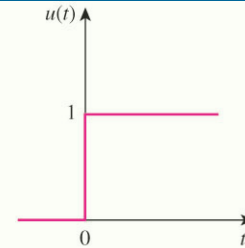


Answer: $i(t) = 2e^{-2t}\text{ A}$

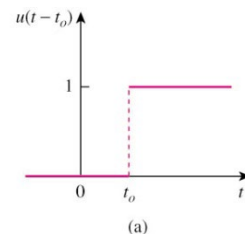
Unit-Step Function

- The **unit step function** $u(t)$ is 0 for negative values of t and 1 for positive values of t .

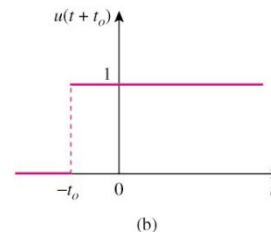
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$



$$u(t - t_o) = \begin{cases} 0, & t < t_o \\ 1, & t > t_o \end{cases}$$



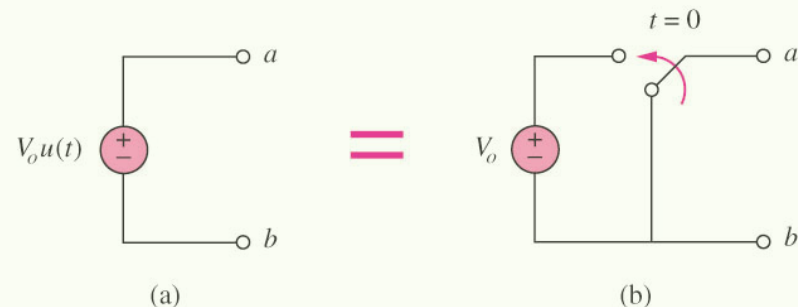
$$u(t + t_o) = \begin{cases} 0, & t < -t_o \\ 1, & t > -t_o \end{cases}$$



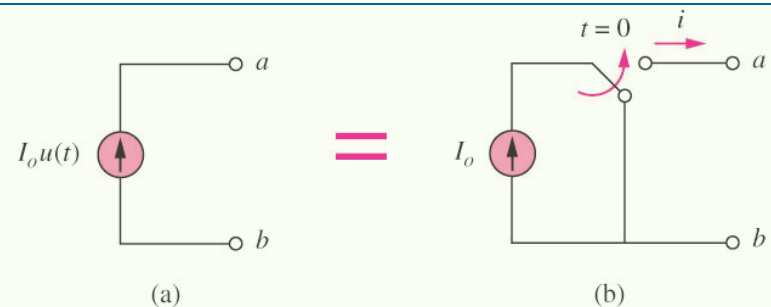
Unit-Step Function

Represent an abrupt change for:

1. voltage source.

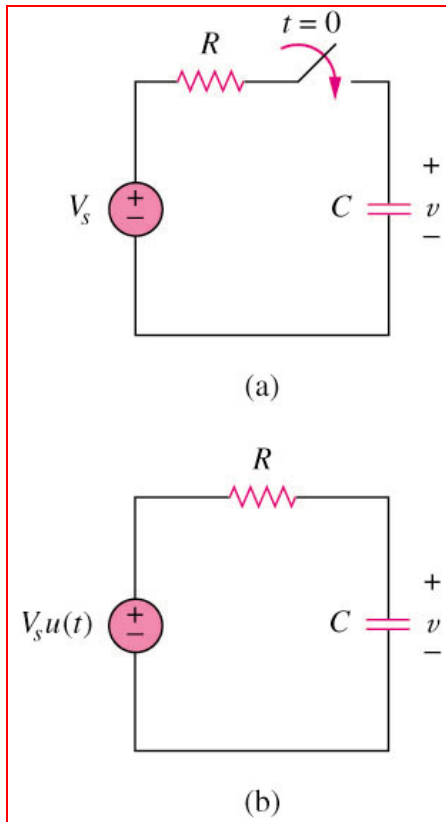


2. for current source:



The Step-Response of a RC Circuit

- The **step response** of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.



- Initial condition:** $v(0-) = v(0+) = V_0$

- Applying KCL,

$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0$$

$$\text{or} \quad \frac{dv}{dt} = -\frac{v - V_s}{RC} u(t)$$

$$v(t) = \begin{cases} V_0 & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau} & t > 0 \end{cases}$$

$$v(t) = v(\infty) + [v(0+) - v(\infty)]e^{-t/\tau}$$

The Step-Response of a RC Circuit

$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0$$

$$\frac{dv}{dt} = -\frac{v - V_s}{RC} u(t)$$

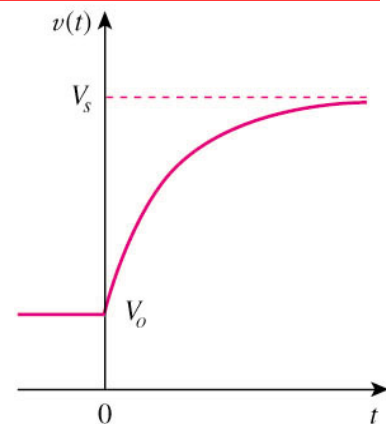
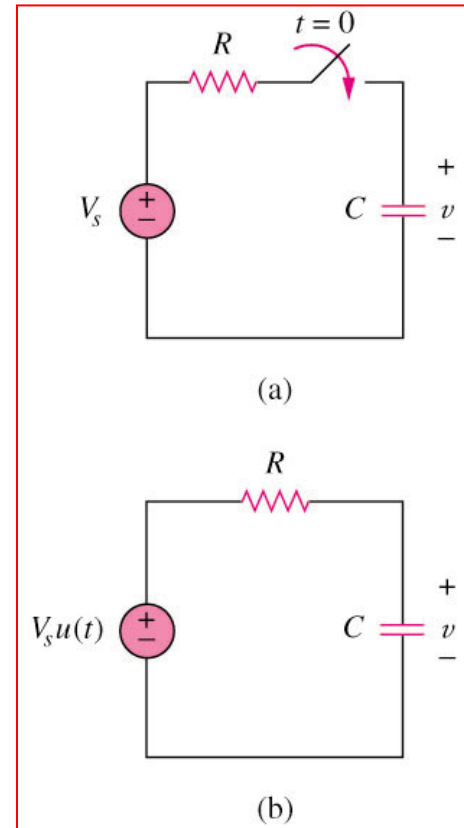
Three steps to find out the step response of an RC circuit:

1. The initial capacitor voltage $v(0)$.
2. The final capacitor voltage $v(\infty)$
DC voltage across C.
3. The time constant $\tau = RC$.

$$v(t) = \begin{cases} V_0 & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau} & t > 0 \end{cases}$$

$$v(t) = v(\infty) + [v(0+) - v(\infty)]e^{-t/\tau}$$

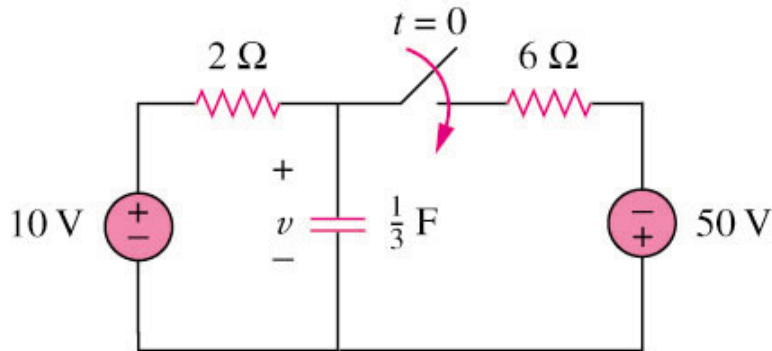
Complete Response =	Natural response (stored energy)	+	Forced Response (independent source)
=	$V_0 e^{-t/\tau}$	+	$V_s (1 - e^{-t/\tau})$



The Step-Response of a RC Circuit

Example 5

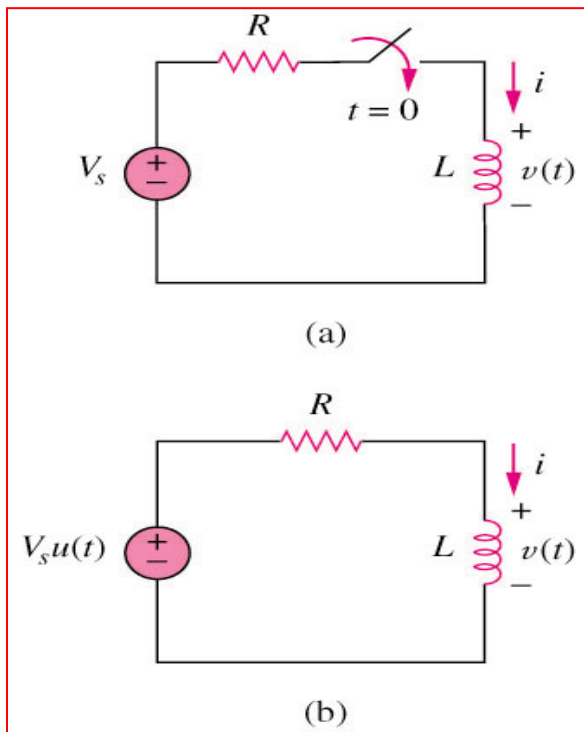
Find $v(t)$ for $t > 0$ in the circuit in below. Assume the switch has been open for a long time and is closed at $t = 0$. Calculate $v(t)$ at $t = 0.5$ sec.



Answer: $v(t) = 15e^{-2t} - 5$ and $v(0.5) = 0.5182V$

The Step-response of a RL Circuit

- The **step response** of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.



- Initial current**
 $i(0^-) = i(0^+) = I_o$
- Final inductor current**
 $i(\infty) = V_s/R$
- Time constant $\tau = L/R$

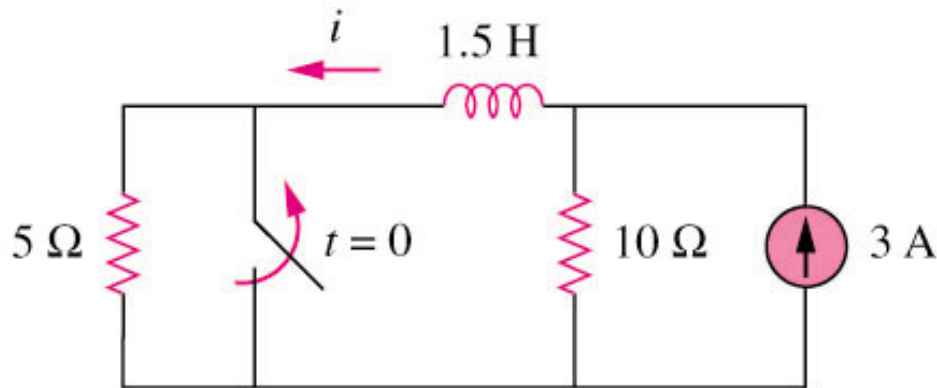
$$i(t) = \frac{V_s}{R} + \left(I_o - \frac{V_s}{R}\right) e^{-\frac{t}{\tau}} u(t)$$

$$i(t) = i(\infty) + [i(0^+) - i(\infty)] e^{-t/\tau}$$

The Step-Response of a RL Circuit

Example 6

The switch in the circuit shown below has been closed for a long time. It opens at $t = 0$. Find $i(t)$ for $t > 0$.



Answer: $i(t) = 2 + e^{-10t}$